

Game-Guided Tactical Corridor Learning for Autonomous Racing Overtakes

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ABSTRACT

Interactive overtaking in autonomous racing requires opponent-aware tactical decisions that remain coherent across repeated replanning and provide a realizable channel for downstream optimization. We address this decision-to-execution problem through game-guided tactical-corridor learning. A robust Stackelberg response-set teacher evaluates near-best opponent responses and supplies action and value supervision to Truncated Quantile Critics. The policy publishes a passing-side intention, time-varying Frenet bounds, and a speed cap. A hysteretic state machine and footprint carver maintain a connected, side-consistent corridor, which a sequential SOCP, speed profiler, and nonlinear MPC realize as closed-loop path–speed motion. Experimental results indicate that the method achieves a 94.7% overtaking success rate and a mean overtaking time of 4.82 s. Success exceeded four matched baselines by 31.2–66.2 percentage points; fixed-corridor and no-FSM ablations reduced success by 18.6 and 12.4 points. These results show that a persistent tactical feasible region provides an effective interface between interactive decisions and closed-loop vehicle execution.

1. Introduction

Autonomous racing combines high speed, narrow maneuvering space, and direct competition with non-cooperative opponents. During an overtake, relative geometry and opponent behavior evolve while the planning loop repeatedly updates its solution. A useful tactical decision must therefore remain coherent long enough to form a continuous passing channel and must expose that channel in a form that the path and speed planners can realize. Full-scale events such as the Abu Dhabi Autonomous Racing League (A2RL) make this requirement practically important because racing software observes competing vehicles and returns drive-by-wire commands under a hard real-time loop [2, 31]. The central problem considered here is the persistence and executability of opponent-aware overtaking decisions under repeated replanning.

1.1. Related work

Traditional overtaking methods organize interaction through hierarchical maneuver planning or explicit dynamic games. In hierarchical architectures, a tactical layer selects among following, defending, and passing behaviors, and a lower layer converts the selected behavior into a path and speed profile. Finite-state and candidate-trajectory planners make this decomposition computationally predictable and allow racing rules, collision checks, and vehicle constraints to be handled by separate modules [5, 6]. Full-scale head-to-head systems retain this modular structure because opponent prediction and local planning must be updated within tight real-time budgets [4, 31]. Thakkar et al. formulated the high-level problem as a discrete racing game that generates target waypoints for either a multi-agent reinforcement-learning controller or a linear-quadratic Nash-game controller [29]. More recently, Toschi et al. constructed opponent-aware drivable tunnels, selected a

tunnel according to the current interaction, and optimized a trajectory within it [34]. These approaches provide interpretable tactical logic and a practical path to real-time implementation. Their passing continuity, however, is principally determined by prescribed state transitions, candidate selection, or tunnel-selection logic; the strategic decision itself is not learned as a persistent feasible set that is maintained across successive planning updates.

Explicit game-theoretic planners model the coupling between competitors more directly. Non-cooperative dynamic games jointly evaluate progress, collision avoidance, blocking, and overtaking, producing behaviors such as shielding, faking, and opportunistic passing that are difficult to obtain from independent trajectory optimization [11, 12]. Iterative linear-quadratic approximations reduce the cost of solving nonlinear multi-player games, but their online burden and local solution quality remain tied to the interaction model and initialization [13]. Hierarchical Stackelberg formulations separate the leader's maneuver choice from the follower's response, while payoff learning adapts the interaction objective from observed behavior [7, 10]. Level- K reasoning offers a bounded model of opponent rationality; trajectory mixing can further hedge against changes in the inferred reasoning level [8]. Constrained-potential and near-potential formulations instead exploit common structure in the players' objectives to reduce the online game-solving burden [14, 15]. Related game formulations have also been used to balance competing human and automation objectives in robust shared control [9]. Collectively, these methods make strategic coupling explicit, yet most publish a maneuver, waypoint sequence, or selected trajectory at each update. Consequently, side commitment and the spatial freedom available to the downstream planner must still be reconstructed by an additional temporal logic layer.

Learning-based methods replace repeated online search with a fixed-cost policy evaluation and can acquire interaction patterns directly from competitive experience. Curriculum learning has produced overtaking policies by gradually increasing opponent difficulty, while large-scale deep reinforcement learning has demonstrated high-performance racing and human-like tactical behavior in simulation [16, 18, 21]. Comparisons with optimal control show that reinforcement learning can exploit complex racing dynamics effectively, although its behavior depends strongly on the training distribution and reward design [22]. To introduce more structure, trajectory-aided policies condition learning on a nominal path, LiDAR-based agents learn directly from local perception, and predictive planners incorporate an opponent forecast into overtaking-trajectory generation [19, 20, 24]. Game-informed deep reinforcement learning couples the policy reward to another vehicle's action, whereas dynamic-graph and multi-agent formulations encode pairwise relations or coordinated behavior among several vehicles [17, 23, 25]. These designs improve interaction awareness, but an end-to-end control action or a single learned trajectory offers little freedom for a constrained planner to reshape the maneuver when the opponent moves during the pass.

Hybrid learning–optimization architectures therefore expose an intermediate decision to a model-based planner. Reiter et al. used reinforcement learning to parameterize the objective of a nonlinear predictive planner, confining exploration to trajectories accepted by the optimizer [30]. In road-traffic applications, learned tactical layers have been used to anticipate disruptive lane changes and to coordinate merging sequences subsequently realized by optimal control [26, 28]. Human feedback provides another source of policy guidance for stabilizing learned driving strategies [27]. Most closely related to a corridor interface, Yang et al. used MARL-generated cooperative reference trajectories to construct spatio-temporal corridors for downstream optimization [33], while Toschi et al. computed alternative drivable tunnels from predicted track occupancy [34]. The present problem differs in the interaction model and in what is learned: the vehicles are non-cooperative, and the policy directly publishes a side-locked feasible region rather than a guiding trajectory or a selector over independently generated tunnels.

The literature thus provides effective game models, learned tactical policies, and corridor-constrained planners, but leaves a specific coupling problem between them. During a close overtake, opponent-aware reasoning must remain consistent across policy updates while preserving sufficient geometric freedom for receding-horizon planning. We address this problem by representing the decision as a persistent tactical feasible region. A robust Stackelberg response-set teacher evaluates projected candidates against near-best opponent responses, TQC distills the resulting action and value supervision into one policy evaluation, and deterministic hybrid logic maintains the selected side while publishing time-varying Frenet bounds and a speed request.

This design transfers strategic interaction reasoning to the planner without reducing the decision to a single path.

1.2. Contributions and paper organization

This paper formulates interactive overtaking as persistent tactical feasible-set generation. The proposed policy learns where and when to commit, preserves the selected side as the encounter evolves, and communicates the decision to the planner through explicit space–speed bounds. A common model-based executor then realizes these bounds as a continuous path–speed reference and closed-loop vehicle motion.

The contributions are threefold.

1. We construct a robust Stackelberg response-set teacher that evaluates projected tactical candidates against near-best opponent responses. Its response-set-optimal action prior and action-conditioned value labels guide TQC toward overtaking opportunities that retain progress and clearance across plausible interaction outcomes.
2. We introduce a planner-compatible tactical representation that combines a learned time-varying corridor with hysteretic side commitment, deterministic opponent-footprint carving, and reference-level tire-admissibility projection. The resulting interface maintains a connected passing region and consistent side choice while the relative geometry changes.
3. We evaluate decision persistence and planner executability in a target-hardware HIL loop. Matched baselines and ablations isolate how corridor shaping, side commitment, game supervision, and distributional learning affect overtaking success and closed-loop feasibility.

Section 2 formulates the overtaking problem and vehicle–track models. Section 3 presents the game-guided corridor policy. Section 4 summarizes the common corridor-constrained reference-generation and closed-loop tracking layer. Section 5 reports the HIL evaluation, comparative analysis, and validation scope.

2. Problem description and modeling

Figure 1 shows the two-vehicle overtaking scenario considered in this study. An ego race car must pass a non-cooperative opponent on a closed track by selecting a persistent passing side, Frenet corridor, and speed request from the vehicle states and local track geometry. The following models define the vehicle and interaction quantities used by the tactical policy and executor.

For vehicle $i \in \{e, o\}$, (X_i, Y_i, ψ_i) denotes the inertial pose and $(v_{x,i}, v_{y,i}, r_i)$ the body-frame velocity and yaw rate, with $V_i = \sqrt{v_{x,i}^2 + v_{y,i}^2}$ and $\beta_i = \text{atan2}(v_{y,i}, v_{x,i})$. Let $\mathcal{W} = \{\text{fl}, \text{fr}, \text{rl}, \text{rr}\}$, and let $\mathbf{r}_q = [x_q, y_q]^\top$ and δ_q denote the position and steering angle of wheel q . The front wheels share the commanded steering angle and the rear wheels are

unsteered. The planar four-wheel dynamics are

$$\begin{bmatrix} \dot{X}_i \\ \dot{Y}_i \\ \dot{\psi}_i \\ \dot{v}_{x,i} \\ \dot{v}_{y,i} \\ \dot{r}_i \end{bmatrix} = \begin{bmatrix} v_{x,i} \cos \psi_i - v_{y,i} \sin \psi_i \\ v_{x,i} \sin \psi_i + v_{y,i} \cos \psi_i \\ r_i \\ \frac{1}{m} \sum_{q \in \mathcal{W}} (F_{x,q} \cos \delta_q - F_{y,q} \sin \delta_q) + r_i v_{y,i} \\ -\frac{F_{\text{drag}}(V_i)}{m} \\ \frac{1}{m} \sum_{q \in \mathcal{W}} (F_{x,q} \sin \delta_q + F_{y,q} \cos \delta_q) - r_i v_{x,i} \\ \frac{1}{I_z} \sum_{q \in \mathcal{W}} [x_q (F_{x,q} \sin \delta_q + F_{y,q} \cos \delta_q) \\ - y_q (F_{x,q} \cos \delta_q - F_{y,q} \sin \delta_q)] \end{bmatrix}. \quad (1)$$

where m and I_z are the vehicle mass and yaw inertia, and F_{drag} is the aerodynamic drag force. Pure-slip tire forces and the resulting quasi-steady-state acceleration envelope are described by

$$\begin{aligned} \zeta_{d,q} &= B_d \chi_{d,q} - E_d (B_d \chi_{d,q} - \arctan(B_d \chi_{d,q})), \\ F_{d,q}(\chi_{d,q}, F_{z,q}) &= D_d(F_{z,q}) \sin(C_d \arctan \zeta_{d,q}), \\ D_d(F_{z,q}) &= F_{z,q} \left[\mu_d + s_d \frac{F_{z,q} - F_{z,0}}{F_{z,0}} \right], \\ B_d &= \frac{\tan(\pi/(2C_d))}{q_{d,\text{pk}}}, \quad d \in \{x, y\}, q \in \mathcal{W}, \\ \begin{bmatrix} a_T \\ a_N \end{bmatrix} &= \frac{1}{m} \sum_{q \in \mathcal{W}} \mathbf{R}(\delta_q - \beta_i) \begin{bmatrix} F_{x,q} \\ F_{y,q} \end{bmatrix}, \\ \mathbf{R}(\varphi) &= \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}, \\ \mathcal{A}_{\text{tire}}(V) &= \{(a_T, a_N) : \Phi_{\text{tire}}(a_T, a_N, V) \leq 1\}. \end{aligned} \quad (2)$$

Here $\chi_{x,q} = \lambda_q$, $\chi_{y,q} = \alpha_q$, and $E_d = 0$ for the fitted curves. Following [35], the envelope is stored as a speed-interpolated map whose traction and braking limits are $a_{T,\text{env}}^+(a_N, V)$ and $b_{T,\text{env}}^-(a_N, V)$.

Let $\mathbf{r}_c(s)$ denote the track centerline parameterized by arc length $s \in [0, L_{\text{trk}})$, with tangent angle $\theta(s)$ and curvature $\kappa(s) = d\theta/ds$. The Frenet geometry and kinematics are

$$\begin{aligned} \mathbf{t}(s) &= [\cos \theta(s), \sin \theta(s)]^\top, \\ \mathbf{n}_c(s) &= [-\sin \theta(s), \cos \theta(s)]^\top, \\ \mathbf{p}_i &= [X_i, Y_i]^\top = \mathbf{r}_c(s_i) + n_i \mathbf{n}_c(s_i), \quad \xi_i = \psi_i - \theta(s_i), \\ v_{s,i} &= V_i \cos(\xi_i + \beta_i), \quad v_{n,i} = V_i \sin(\xi_i + \beta_i), \\ \begin{bmatrix} \dot{s}_i \\ \dot{n}_i \\ \dot{\xi}_i \end{bmatrix} &= \begin{bmatrix} v_{s,i}/(1 - \kappa(s_i)n_i) \\ v_{n,i} \\ r_i - \kappa(s_i)v_{s,i}/(1 - \kappa(s_i)n_i) \end{bmatrix}. \end{aligned} \quad (3)$$

The lateral coordinate is positive to the left, with $n_{\text{trk}}^-(s) \leq n \leq n_{\text{trk}}^+(s)$. The spatial model is used within $1 - \kappa n \geq \varepsilon_s$, $|\xi| \leq \xi_{\text{max}} < \pi/2$, and $v_s \geq \underline{v}_s > 0$; Table 1 lists the limits.

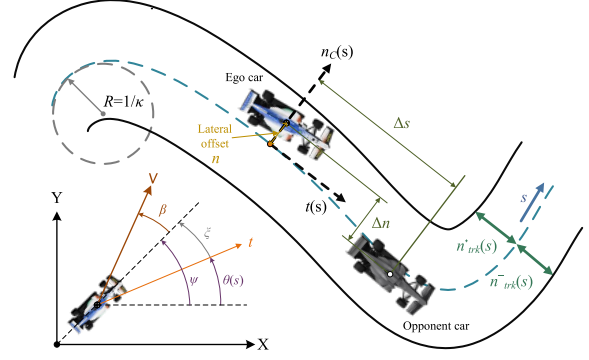


Figure 1: Vehicle motion and relative geometry in the Frenet frame.

Centerline projection, unwrapped at the start–finish line, gives the relative ego–opponent quantities

$$\begin{aligned} g &= \Delta s = ((s_o - s_e + L_{\text{trk}}/2) \bmod L_{\text{trk}}) - L_{\text{trk}}/2, \\ \Delta n &= n_o - n_e, \quad \Delta v = v_{s,e} - v_{s,o}. \end{aligned} \quad (4)$$

Here $g > 0$ denotes an opponent ahead and $\Delta v > 0$ denotes the ego closing; these variables describe the duel geometry used by the tactical policy.

For spatial path generation, s is the independent variable, $\mathbf{z}(s) = [n(s), \xi(s)]^\top$ is the geometric state, and $\omega(s)$ is the path curvature. The resulting model is

$$\mathbf{z}'(s) = \mathbf{f}_s(\mathbf{z}, \omega; s) = \begin{bmatrix} (1 - \kappa n) \tan \xi \\ (1 - \kappa n) \omega / \cos \xi - \kappa \end{bmatrix}. \quad (5)$$

Section 4 uses its first-order approximation in the sequential SOCP.

3. Game-Guided Tactical Corridor Policy

The tactical layer converts opponent-aware interaction reasoning into a persistent feasible region. Response-set supervision identifies overtaking opportunities across plausible opponent reactions, TQC amortizes this reasoning into a real-time policy, and the hybrid publisher maintains the selected side while constructing explicit corridor bounds for the planner.

3.1. Robust Stackelberg response-set supervision

At each tactical update, the encoder combines the two vehicle states, local track preview, current mode, side lock, and execution health into $\tilde{\mathbf{o}}_t$. The actor produces

$$\mathbf{a}^{\text{raw}} = [L^{\text{cap}}, R^{\text{cap}}, d_{\text{safe}}, V_{\text{cap}}^{\text{req}}, g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}, b_{\text{lat}}, w_{\text{safe}}, q_{\text{side}}]^\top. \quad (6)$$

where the components specify the usable track widths, footprint margin, requested speed cap, ordered interaction distances, defensive interval, and side intention. Together, these variables describe the spatial extent, timing, and side commitment of the passing opportunity. The Euclidean projection $\Pi_{\mathcal{A}}$ enforces the component bounds in Table 2, the guard order $g_{\text{ot}} + \Delta_g \leq g_{\text{chase}} \leq g_{\text{abort}} - \Delta_g$, and the minimum

corridor width. The tactical action therefore follows the single chain

$$\mathbf{a}^{\text{raw}} \xrightarrow{\Pi_A} \mathbf{a}^{\text{B}} \xrightarrow{\Pi_{\text{tire}}} \mathbf{a}^{\text{F}}, \quad \mathcal{G}_t = (m_{t+1}, \sigma_{t+1}^{\text{lock}}, V_{\text{cap},t}^{\text{F}}, C_t).$$

The neural policy determines the tactical sample; mode, lock, corridor, and speed-cap publication are resolved by Algorithm 1.

The offline teacher forms deterministic leader and follower banks by combining the three side codes with a Sobol design over the remaining action components. Every candidate is passed through the same bound projection, corridor carver, and tire screen used online; duplicate or infeasible candidates are discarded. Let $s_t^G = (\tilde{o}_t, \chi_t^G)$ denote the teacher state, where χ_t^G collects the publisher, executor, and one-step reward memory carried between tactical updates, and let $\mathcal{A}_i^G(s_t^G)$ denote the candidate bank for role i . Each candidate pair induces the coupled executable preview

$$(\mathcal{R}_e, \mathcal{R}_o) = \mathcal{P}_{N_{\text{sta}}}(s_t^G; \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}), \quad (7)$$

where $\mathcal{P}_{N_{\text{sta}}}$ recursively applies the stateful publisher in Algorithm 1 and the common execution stack. The candidate pair remains fixed over the preview while both vehicle and hybrid states evolve jointly; both utilities use the corresponding role-relative features of this preview.

Let $Z_g = \sum_{k=0}^{N_{\text{sta}}-1} \gamma_g^k$ and $c_w = \sum_{j=1}^{10} [-w_j]_+$. The normalized role-relative utility is

$$U_i(s_t^G, \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}) = \frac{1}{Z_g \|\mathbf{w}\|_1} \sum_{k=0}^{N_{\text{sta}}-1} \gamma_g^k [\mathbf{w}^\top \boldsymbol{\phi}_{i,k}(\mathcal{R}_e, \mathcal{R}_o) + c_w], \quad (8)$$

where $\boldsymbol{\phi}_i \in [0, 1]^{10}$ contains the role-relative reward features in A. Hence $U_i \in [0, 1]$, and ε_{BR} directly specifies the admissible follower-utility gap.

$$\begin{aligned} U_o^{\text{max}}(s_t^G, \mathbf{a}_e^{\text{F}}) &= \max_{\mathbf{a}_o^{\text{F}} \in \mathcal{A}_o^G(s_t^G)} U_o(s_t^G, \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}), \\ \mathfrak{R}_o^\varepsilon(s_t^G, \mathbf{a}_e^{\text{F}}) &= \{\mathbf{a}_o^{\text{F}} \in \mathcal{A}_o^G(s_t^G) : \\ &\quad U_o^{\text{max}}(s_t^G, \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}) - U_o(s_t^G, \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}) \leq \varepsilon_{\text{BR}}\}, \\ \mathcal{V}_g(s_t^G, \mathbf{a}_e^{\text{F}}) &= \min_{\mathbf{a}_o^{\text{F}} \in \mathfrak{R}_o^\varepsilon(s_t^G, \mathbf{a}_e^{\text{F}})} U_e(s_t^G, \mathbf{a}_e^{\text{F}}, \mathbf{a}_o^{\text{F}}), \\ \mathbf{a}_{\text{th}}^*(s_t^G) &\in \arg \max_{\mathbf{a}_e^{\text{F}} \in \mathcal{A}_e^G(s_t^G)} \mathcal{V}_g(s_t^G, \mathbf{a}_e^{\text{F}}), \\ \mathcal{D}_g &= \left\{ (\tilde{o}_n, \mathbf{a}_{g,n}^{\text{F}}, v_{g,n}, \right. \\ &\quad \left. \mathbf{a}_{\text{th},n}^*) : v_{g,n} = \mathcal{V}_g(s_n^G, \mathbf{a}_{g,n}^{\text{F}}) \right\}. \end{aligned} \quad (9)$$

Here $\varepsilon_{\text{BR}} = 0.05$ retains responses within 0.05 of the best candidate on the normalized utility scale.

Before publication, the requested speed cap is certified against the speed-dependent tire envelope on the provisional corridor midpoint. For each trial $v \in \mathcal{V}_s(V_{\text{cap}}^{\text{req}})$, define the envelope-unclipped kinetic-energy template

$$\lambda_k = \frac{\sum_{j=0}^{k-1} \Delta \ell_j}{\sum_{j=0}^{N_{\text{sta}}-2} \Delta \ell_j}, \quad \hat{V}_k(v) = \sqrt{V_e^2 + \lambda_k(v^2 - V_e^2)}.$$

Using the midpoint curvature $\hat{\kappa}_k$, the induced demand and total projection are

$$\begin{aligned} \hat{a}_{T,k}(v) &= \frac{\hat{V}_{k+1}^2(v) - \hat{V}_k^2(v)}{2\Delta \ell_k} + \frac{F_{\text{drag}}(\hat{V}_k(v))}{m}, \\ \hat{a}_{N,k}(v) &= \hat{\kappa}_k \hat{V}_k^2(v), \\ \hat{\rho}(v) &= \max_{0 \leq k \leq N_{\text{sta}}-2} \Phi_{\text{tire}}(\hat{a}_{T,k}(v), \hat{a}_{N,k}(v), \hat{V}_k(v)), \\ \mathcal{V}_{\text{adm}}(\mathbf{a}) &= \{v \in \mathcal{V}_s(V_{\text{cap}}^{\text{req}}) : \hat{\rho}(v) \leq 1\}, \\ \Pi_{\text{tire}}(\mathbf{a}; \tilde{o}) &= \begin{cases} \mathbf{a}[\max \mathcal{V}_{\text{adm}}(\mathbf{a})], & \mathcal{V}_{\text{adm}}(\mathbf{a}) \neq \emptyset, \\ \mathbf{a}_{\text{HF}}^{\text{F}}(\tilde{o}), & \mathcal{V}_{\text{adm}}(\mathbf{a}) = \emptyset. \end{cases} \end{aligned} \quad (10)$$

Here V_e is the current ego speed, $\mathcal{V}_s$ is the uniformly sampled finite grid from V_{min} to $V_{\text{cap}}^{\text{req}}$, and $\mathbf{a}[v]$ changes only the cap component. The total branch $\mathbf{a}_{\text{HF}}^{\text{F}}$ performs the current-state HOLD reconstruction and returns the braking-fallback contract if its repeated search remains empty.

3.2. Game-guided distributional policy

The real-time policy maps the evolving duel geometry to the tactical feasible set at a fixed computational cost. Its 51-dimensional observation concatenates 27 duel-state features with two 12-dimensional track summaries centred at the ego and opponent. The duel features contain normalized Frenet states and accelerations, relative progress, lateral gap and closing rate, time to contact, closing-speed code, side lock, and a nine-state race-mode code. Each track summary reports mean curvature, peak absolute curvature, and minimum width over four consecutive bins.

The environment reward is

$$\begin{aligned} r_t &= \mathbf{w}^\top \boldsymbol{\phi}_t, \\ \boldsymbol{\phi}_t &= [\phi_{\text{ot}}, \phi_{\text{ovb}}, \phi_{\text{col}}, \phi_{\text{lat}}, \phi_{\text{crush}}, \phi_{\text{app}}, \\ &\quad \phi_{\text{sm}}, \phi_{\text{gs}}, \phi_{\text{sbs}}, \phi_{\text{wide}}]^\top. \end{aligned} \quad (12)$$

A gives the normalized definitions and weights. The same role-swapped features define Eq. (8), aligning the teacher and online reward at the level of overtaking progress, clearance, smoothness, and usable corridor width.

Projected-action TQC. Truncated Quantile Critics (TQC) [32] discard the largest target atoms before the quantile-Huber backup. Environment replay $\mathcal{D}$ stores the final tactical action $\mathbf{a}_t^{\text{F}}$. For a raw target-policy sample, the projected action and truncated target are

$$\begin{aligned} \mathbf{a}_{t+1}^{\text{F}} &= \Pi_{\text{tire}}(\Pi_A(\mathbf{a}_{t+1}^{\text{raw}}); \tilde{o}_{t+1}), \\ \ell_{t+1} &= \log \pi_\theta(\mathbf{a}_{t+1}^{\text{raw}} | \tilde{o}_{t+1}), \\ y_{t,k} &= r_t + \gamma(1 - d_t) \left[z'_{(k)}(\tilde{o}_{t+1}, \mathbf{a}_{t+1}^{\text{F}}) - \alpha \ell_{t+1} \right], \\ \mathcal{Y}_t &= \text{sg}(\{y_{t,k}\}_{k=1}^K), \quad K < 2N_q. \end{aligned} \quad (13)$$

Thus target critics and replay share the same projected-action support, while entropy remains the density of the raw stochastic policy. Actor and prior updates pass gradients

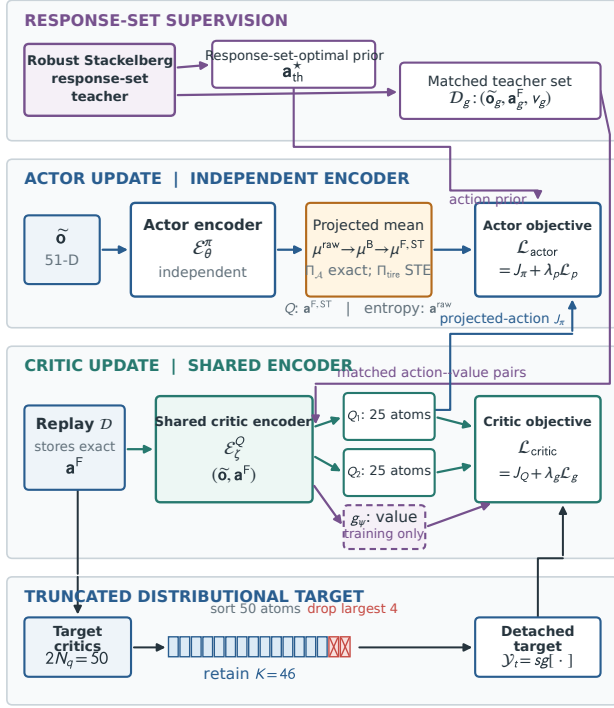


Figure 2: Response-set-guided projected-action TQC training. The actor uses an independent encoder and receives the response-set-optimal action prior. The two quantile heads and the training-only robust-value head share the critic encoder, while matched samples in D_g supervise their corresponding projected actions. The largest four target atoms are discarded before the detached critic backup.

through the non-smooth tire search using

$$\mathbf{a}^{F,ST} = \mathbf{a}^B + \text{sg}[\mathbf{a}^F - \mathbf{a}^B]. \quad (14)$$

This projected-action surrogate uses the exact projected action forward and an identity backward map through Π_{tire} .

Two quantile heads share $h_\zeta = \mathcal{E}_\zeta^O(\tilde{\mathbf{o}}, \mathbf{a}^F)$. For matched $(\tilde{\mathbf{o}}_g, \mathbf{a}_g^F, v_g, \mathbf{a}_{th}^*) \sim D_g$, let $h_{\zeta,g} = \mathcal{E}_\zeta^O(\tilde{\mathbf{o}}_g, \mathbf{a}_g^F)$; the updates are

$$\begin{aligned} \mathcal{L}_{\text{critic}} &= J_Q + \lambda_g \mathbb{E}_{D_g} \left[(g_\psi(h_{\zeta,g}) - N_V(v_g))^2 \right], \\ \mathcal{L}_{\text{actor}} &= J_\pi + \lambda_p \mathbb{E}_{D_g} \left[\|N_a(\mu_\theta^F) - N_a(\mathbf{a}_{th}^*)\|_2^2 \right]. \end{aligned} \quad (15)$$

Here J_Q and J_π are the standard TQC critic and entropy-regularized actor losses; N_a and N_V normalize actions and teacher values. The actor encoder is independent, whereas the two quantile heads and the training-only robust-value head share the critic encoder. Consequently, D supplies Bellman transitions and D_g supplies only matched action-value supervision. Figure 2 summarizes these training and deployment paths. The action prior guides corridor placement, while the auxiliary value labels organize the critic representation according to the robustness of the corresponding passing region.

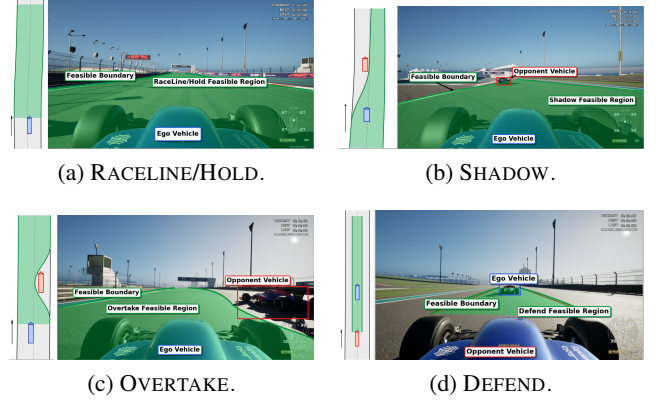


Figure 3: Nominal geometry of the persistent tactical modes. Each panel pairs the Frenet feasible region with the corresponding simulator view; the shaded interval is the region delivered to the planner.

3.3. Persistent intention and corridor synthesis

The state machine carries the selected passing opportunity across successive tactical updates. This temporal persistence complements the spatial corridor: the policy may refine its width and speed request as the encounter evolves, while the committed side remains consistent through the pass. Let g_t and Δv_t be the sampled relative quantities defined in Eq. (4). A feasible side is committed as the ego closes on the opponent, retained through SHADOW and OVERTAKE, and released only after a confirmed pass, an abort, or an execution-health failure. Health and abort take precedence over passing, completed passes take precedence over continued overtaking, and an unlocked ego enters DEFEND only when a rear opponent is closing. The post-pass indicator P_t used in the reward denotes N_{post} consecutive updates beyond the overtake clearance threshold.

The geometric operation that realizes the selected side is the corridor carver. For $B_{\text{cap},k} = [\ell_k, u_k]$, the vehicle footprint and one-sided exclusion are

$$\begin{aligned} \rho_{\text{exc},k} &= d_{\text{safe}} + \frac{1}{2} \sum_{j \in \{e,o\}} (L_j |\sin \xi_{j,k}| + W_j |\cos \xi_{j,k}|), \\ B_{\text{cap},k} &= [\max\{n_{\text{tr},k}^-, -R^{\text{cap}}\}, \min\{n_{\text{tr},k}^+, L^{\text{cap}}\}], \\ C_k^{\text{side}}(\sigma, \eta) &= \begin{cases} [\ell_k + \eta[n_{o,k} + \rho_{\text{exc},k} - \ell_k]_+, u_k], & \sigma = +1, \\ [\ell_k, u_k - \eta[u_k - n_{o,k} + \rho_{\text{exc},k}]_+], & \sigma = -1. \end{cases} \end{aligned} \quad (16)$$

Here $\sigma = +1$ and -1 select the left and right passing sides, respectively. SHADOW uses $\eta = \text{clip}((g_{\text{chase}} - \hat{g}_k)/(g_{\text{chase}} - g_{\text{ol}}), 0, 1)$, whereas OVERTAKE uses full exclusion ($\eta = 1$) on the locked side. RACELINE uses the capped track interval; DEFEND intersects it with a width- w_{safe} interval centred at the clipped offset $n_{o,k} + b_{\text{lat}}$; and HOLD retains only the connected component containing the ego vehicle. Figure 3 illustrates these geometric semantics.

Algorithm 1 summarizes the complete route from off-line interaction supervision to online corridor publication. Immediate contraction and smoothed relaxation allow the

Algorithm 1 Game-supervised tactical-corridor policy

Require: teacher states and replay $\mathcal{D}$; online states, track preview, previous $(m, \sigma^{\text{lock}}, C)$, and execution health

- 1: **Offline teacher and training:**
- 2: **for** each teacher state **do**
- 3: combine Sobol samples with left/neutral/right codes; apply $\mathbf{a}^{\text{raw}} \rightarrow \mathbf{a}^{\text{B}} \rightarrow \mathbf{a}^{\text{F}}$
- 4: discard duplicate or infeasible candidates; generate the coupled preview (7) for each candidate pair
- 5: evaluate (8), retain near-best follower responses, and compute (9)
- 6: store every matched $(\tilde{\mathbf{o}}, \mathbf{a}^{\text{F}}, v_g, \mathbf{a}_{\text{th}}^*)$ in $\mathcal{D}_g$
- 7: **end for**
- 8: update TQC from $\mathcal{D}$ and the prior/value terms from $\mathcal{D}_g$ using Eq. (15)
- 9: **Online tactical publication:**
- 10: sample $\mathbf{a}^{\text{raw}}$, apply $\Pi_{\mathcal{A}}$, predict the opponent at constant Frenet velocity, and evaluate the available side widths
- 11: **if** h_i is invalid, or a locked state satisfies $g_i \geq g_{\text{abort}}$ or loses $L_{\text{min}}^{\text{corr}}$ width **then**
- 12: set HOLD and release the side lock
- 13: **else if** the locked ego satisfies $g_i \leq -g_{\text{ot}}$ for N_{post} updates **then**
- 14: set $P_i = 1$, return to RACELINE, and release the lock
- 15: **else if** a lock exists, or a non-neutral feasible side is requested with $0 < g_i \leq g_{\text{chase}}$ and $\Delta v_i \geq 0$ **then**
- 16: retain/commit that side; select SHADOW for $g_i > g_{\text{ot}}$ and OVERTAKE otherwise
- 17: **else if** $\sigma^{\text{lock}} = 0$, $-g_{\text{chase}} \leq g_i < 0$, and $\Delta v_i < 0$ **then**
- 18: select DEFEND
- 19: **else**
- 20: select RACELINE
- 21: **end if**
- 22: construct the mode interval using (16) and the corresponding raceline, defend, or connected-hold rule; use $\eta = 1$ in OVERTAKE and the gap-based ramp in SHADOW
- 23: evaluate (11), including its deterministic current-state HOLD/braking branch
- 24: apply immediate contraction and smoothed relaxation; on width failure, revalidate the preceding corridor for at most one update
- 25: **return** $(m, \sigma^{\text{lock}}, V_{\text{cap}}^{\text{F}}, C)$ if valid; otherwise return braking-fallback status

feasible region to respond quickly to reduced space while expanding progressively as the passing channel opens. The preceding corridor is revalidated for one update when a new interval loses the required width.

4. Corridor-Constrained Planning and Reference Tracking

The execution layer realizes the persistent tactical feasible region as a continuous path–speed reference and actuator commands. The published corridor defines the spatial optimization domain, while the speed request sets the tactical pace throughout the maneuver. At station s_k , $[n_k, \bar{n}_k]$ is the footprint-eroded vehicle-center interval. Sequential iteration j linearizes Eq. (5) as $\hat{\mathbf{f}}_{s,k}^j = \mathbf{A}_k^j \mathbf{z}_k + \mathbf{B}_k^j \omega_k + \mathbf{b}_k^j$ and solves one corridor-constrained SOCP subproblem:

$$\begin{aligned} \mathbf{y}^{j+1} = \arg \min_{\mathbf{y}} \quad & w_{\omega} \sigma_k^{\omega} + w_{n,T} \epsilon_n + w_{\xi,T} \epsilon_{\xi} \\ \text{s.t. } \quad & \mathbf{z}_0 = [n_e, \xi_e]^{\top}, \quad \mathbf{z}_{k+1} - \mathbf{z}_k = \frac{\Delta s_{\text{sta}}}{2} (\hat{\mathbf{f}}_{s,k}^j + \hat{\mathbf{f}}_{s,k+1}^j), \\ & n_k \leq n_k \leq \bar{n}_k, \quad 1 - \kappa(s_k) n_k \geq \epsilon_s, \\ & |\xi_k| \leq \xi_{\text{max}}, \quad |\omega_k| \leq \omega_{\text{max}}, \\ & \|[2\omega_k, \sigma_k^{\omega} - 1]^{\top}\|_2 \leq \sigma_k^{\omega} + 1, \\ & |n_{N-1} - n_T| \leq \epsilon_n, \quad |\xi_{N-1} - \xi_T| \leq \epsilon_{\xi}, \quad \epsilon_n, \epsilon_{\xi} \geq 0. \end{aligned} \quad (17)$$

Here $\mathbf{y} = \{\mathbf{z}_k, \omega_k, \sigma_k^{\omega}, \epsilon_n, \epsilon_{\xi}\}$, $\mathbf{z}_k = [n_k, \xi_k]^{\top}$, and $\omega_{\text{max}} = \tan \delta_{\text{max}} / (l_f + l_r)$. The terminal target is the clipped raceline in RACELINE and the corridor midpoint otherwise, with $\xi_T = 0$. The conic epigraph penalizes path curvature, and the terminal slacks soften convergence to $(n_T, 0)$. The nonlinear Frenet model is re-evaluated after each sequential update.

The forward–backward sweep distributes the certified tactical cap over the optimized path. For curvature $\kappa_{p,k}$, the envelope supplies the traction and braking limits $a_{T,k}^+$ and $b_{T,k}^-$ at $(\kappa_{p,k} V^2, V)$, after subtracting the resistance acceleration in traction and adding it in braking. The steady tactical request is first converted into a braking-reachable stationwise cap $V_{\text{cap},k}^{\text{P}}$. Intersecting it with the largest envelope-admissible curvature speed gives $V_{\text{lim},k} = \min\{\bar{V}_{\text{curv},k}, V_{\text{cap},k}^{\text{P}}\}$. The resulting forward–backward sweep is

$$\begin{aligned} V_0^f &= V_e, \\ V_{k+1}^f &= \text{clip}_{[V_{\text{min}}, V_{\text{lim},k+1}]} \sqrt{(V_k^f)^2 + 2\Delta \ell_k a_{T,k}^+ (V_k^f)}, \\ V_{N-1}^r &= \min\{V_{N-1}^f, V_{\text{lim},N-1}\}, \\ V_k^r &= \min\left\{V_k^f, V_{\text{lim},k}, \sqrt{(V_{k+1}^r)^2 + 2\Delta \ell_k b_{T,k+1}^- (V_{k+1}^r)}\right\}. \end{aligned} \quad (18)$$

The forward pass starts at the measured speed, whereas the backward pass enforces downstream braking reachability. Trapezoidal integration of $1/V_k^r$ provides the time stamps used to resample the path–speed reference on the controller grid.

The RC-NMPC tracks the resampled path–speed reference on a fixed prediction grid. Its state and input are $\chi_k = [x_k, y_k, \theta_k]^{\top}$ and $\mathbf{v}_k = [V_k, \delta_k]^{\top}$, respectively. The controller solves

$$\begin{aligned} \min_{\{\mathbf{v}_k\}} \quad & q_{\Delta\delta} (\delta_0 - \delta_{-1})^2 + \sum_{k=0}^{N_c-1} \left(q_c e_{c,k+1}^2 + q_{\psi} e_{\psi,k+1}^2 + e_{v,k}^{\top} \mathbf{R}_v e_{v,k} \right) \\ \text{s.t. } \quad & \chi_0 = \mathbf{0}, \quad \chi_{k+1} = \chi_k + T_c \begin{bmatrix} V_k \cos \theta_k \\ V_k \sin \theta_k \\ V_k \tan \delta_k / (l_f + l_r) \end{bmatrix}, \quad (19) \\ & V_{\text{min}} \leq V_k \leq \min\{v_{x,\text{max}}, V_{\text{cap},k}^{\text{C}}\}, \\ & |\delta_k| \leq \delta_{\text{max}}, \quad |V_k - V_{k-1}| \leq \Delta V_{\text{max}}, \\ & |\delta_k - \delta_{k-1}| \leq \dot{\delta}_{\text{max}} T_c. \end{aligned}$$

Here $\mathbf{v}_k^r = [V_k^r, \delta_k^r]^{\top}$, with $\delta_k^r = \arctan((l_f + l_r) \kappa_{p,k}^r)$. The quantities $e_{c,k} = (\mathbf{n}_k^r)^{\top} (\mathbf{p}_k - \mathbf{p}_k^r)$, $e_{\psi,k} = \text{wrap}(\theta_k - \theta_k^r)$, and $e_{v,k} = \mathbf{v}_k - \mathbf{v}_k^r$ are the contouring, heading, and input tracking errors. The stagewise bound $V_{\text{cap},k}^{\text{C}}$ is obtained by rate-limiting the steady tactical request from the previous speed command. Realized tire demand is reported in Section 5.

Algorithm 2 Common reference generator and executor

Require: guidance $\mathcal{G}_t$, current state, previous valid reference

- 1: erode the track by the ego footprint, intersect it with $\mathcal{C}_t$, and choose the mode-dependent terminal target
- 2: initialize the spatial path from the previous valid solution or the corridor midpoint
- 3: **for** each configured sequential-convexification iteration **do**
- 4: solve SOCP (17), update the linearization, and evaluate the nonlinear Frenet and corridor residuals
- 5: stop when the solver and residual tolerances in $\mathbf{A}$ are satisfied
- 6: **end for**
- 7: set valid if the path passes the solver, domain, and residual checks
- 8: **if** valid **then**
- 9: query $\mathcal{A}_{\text{tire}}(V)$ and construct $V_{\text{cap},k}^P$ from V_{cap}^F , measured speed, and available braking
- 10: intersect $V_{\text{cap},k}^P$ with the curvature-speed envelope, run (18), and update valid from the tire check
- 11: **end if**
- 12: **if** valid **then**
- 13: time-resample the path-speed reference and rate-limit $V_{\text{cap},t}^F$ to obtain $V_{\text{cap},t}^C$
- 14: solve RC-NMPC (19) and update valid from its status
- 15: **end if**
- 16: **if** valid **then**
- 17: map $(V_{0|t}^{c*}, \delta_{0|t}^{c*})$ to throttle, brake, steering, and gear
- 18: **else**
- 19: revalidate the previous reference for one cycle; otherwise request current-state HOLD and issue rate-limited braking with the preceding steering angle
- 20: **end if**
- 21: **return** four-channel command and health status

5. Experimental results and discussion**5.1. Experimental platform and configuration**

The Abu Dhabi Autonomous Racing League (A2RL) provides a full-scale head-to-head racing environment in which EAV24 vehicles compete autonomously at the Yas Marina Circuit. We reproduced this operating chain in a target-hardware HIL platform (Fig. 4). The AutoVerse EAV24 plant and track environment ran on a simulation server and generated radar, GNSS/IMU, wheel-speed, and race-control data. These streams reached the Neosys RGS-8805GC onboard computer over a dedicated Ethernet link using DDSI-RTPS over UDP/IP. The tactical, planning, and control modules executed on the target computer, which returned throttle, brake, steering, and gear commands to the simulated vehicle through CAN.

Ego position, velocity, attitude, and acceleration are delivered at 100 Hz with the configured localization noise and latency. Opponent perception is updated at 20 Hz with radar range-noise base/scale values of 0.02/0.003, 0.003-rad azimuth noise, a 0.22 velocity-noise scale, a 10% packet-drop rate, and front/left/right/rear time offsets of 0/40/40/20 ms. The decision and planning modules run at 50 Hz, and the control module outputs throttle, brake, steering, and gear commands at 100 Hz.

Statistical results aggregate 1000 Monte Carlo closed-loop simulations per setting. OSR is the fraction of completed overtakes, and TTO is their completion time. Minimum gap, mean TTC, and contact/track-violation rates describe interaction safety, while PFR is the fraction of feasible planning updates.

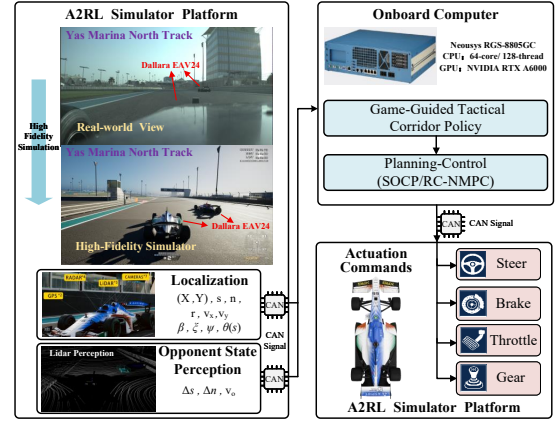


Figure 4: Target-hardware HIL topology. The AutoVerse server publishes radar, GNSS/IMU, wheel-speed, and race-control data to the RGS-8805GC over Ethernet using DDSI-RTPS over UDP/IP; the target computer returns throttle, brake, steering, and gear commands to the EAV24 vehicle model through CAN.

5.2. Overtaking performance under representative track conditions

To examine whether the learned feasible region remains coherent across different overtaking geometries, Fig. 5 presents four encounters at separated locations on the Yas Marina circuit. Each spatial view pairs the published corridor and planned path with synchronized SHADOW and OVERTAKE frames. Cases I and II are then examined in detail as complementary examples of short-window high-speed commitment and persistent passing through a sharp-corner sequence.

Case I: high-speed overtake with a short commitment window

Case I begins near 69 m s^{-1} with a substantially slower opponent and a short commitment window. Figure 6 shows the ordered transition from RACELINE through SHADOW to OVERTAKE. The side-locked corridor forms a continuous lateral channel inside the physical track boundaries, and the optimized ego path follows this channel while establishing separation from the opponent. The actual speed tracks the desired profile and exceeds 70 m s^{-1} during the pass, showing that the tactical opportunity is preserved through high-speed execution.

Figure 7 shows the closed-loop realization of this passing channel. Lateral, speed, and heading errors remain concentrated around zero, with brief transients during the tactical transition, and the complete acceleration sequence remains inside the speed-dependent tire-demand envelope. The spatial decision therefore propagates from side commitment to a trackable path-speed reference throughout the high-speed pass.

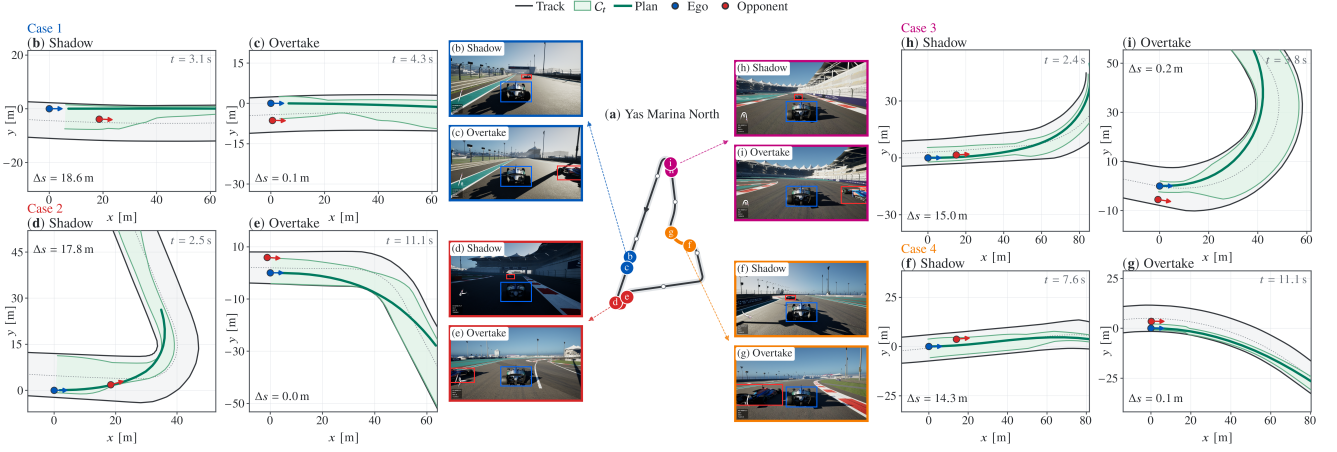


Figure 5: Representative overtaking situations on the Yas Marina circuit. Each colored location links the spatial corridor and planned path to synchronized SHADOW and OVERTAKE simulator views.

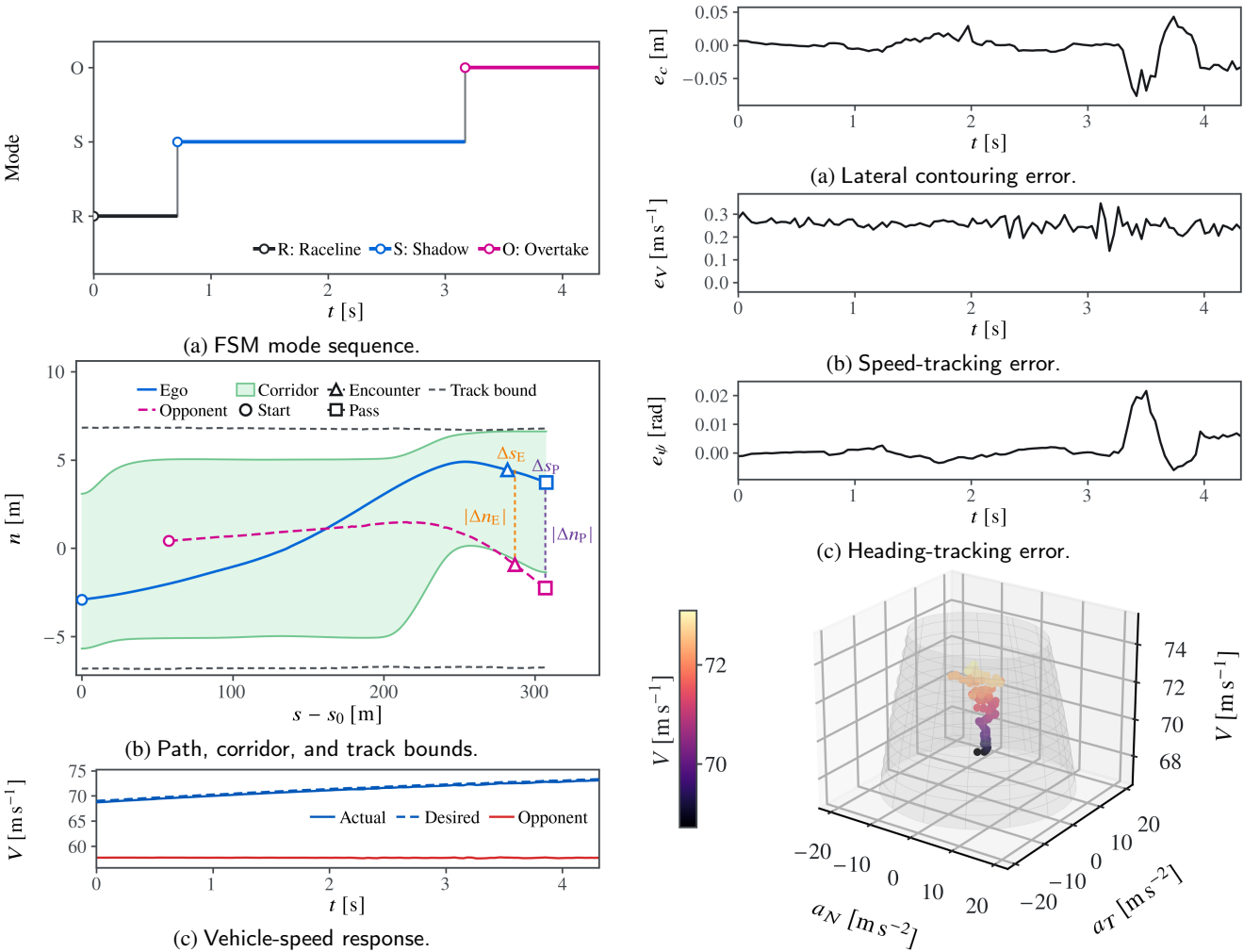


Figure 6: Tactical and speed evolution in Case I. (a) The FSM establishes a persistent passing phase. (b) The planned path remains inside the published corridor and physical track boundaries. (c) The actual speed tracks the desired profile above 70 m s⁻¹ while closing on the opponent.

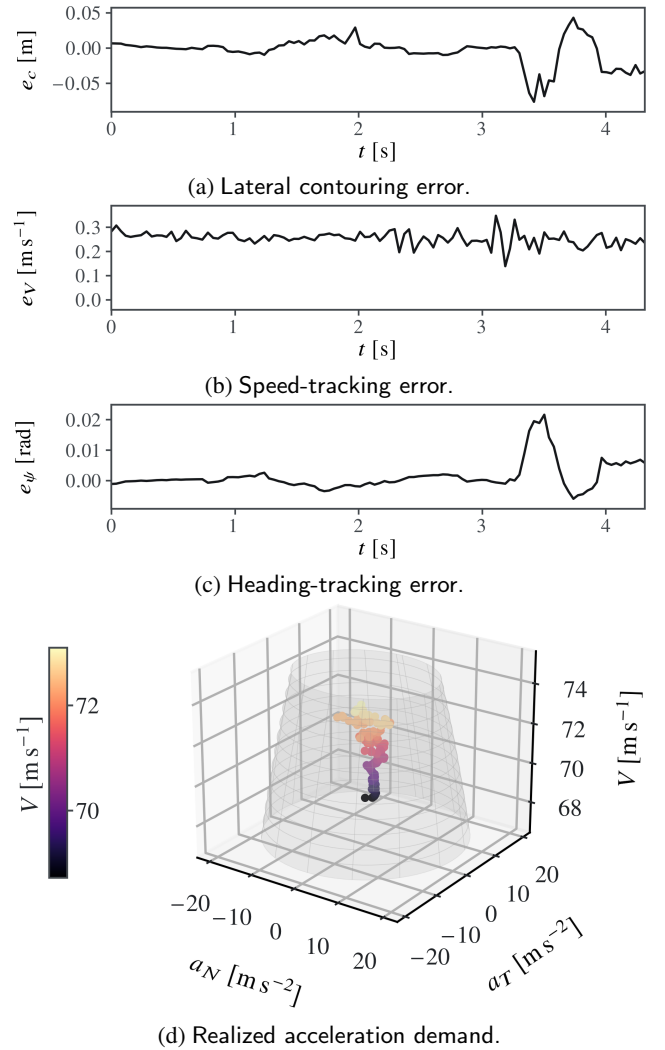


Figure 7: Closed-loop tracking and tire-demand response in Case I. (a–c) Lateral, speed, and heading errors remain bounded through the tactical transition and pass. (d) All recorded tangential–normal acceleration samples remain inside the declared speed-dependent tire-demand envelope.

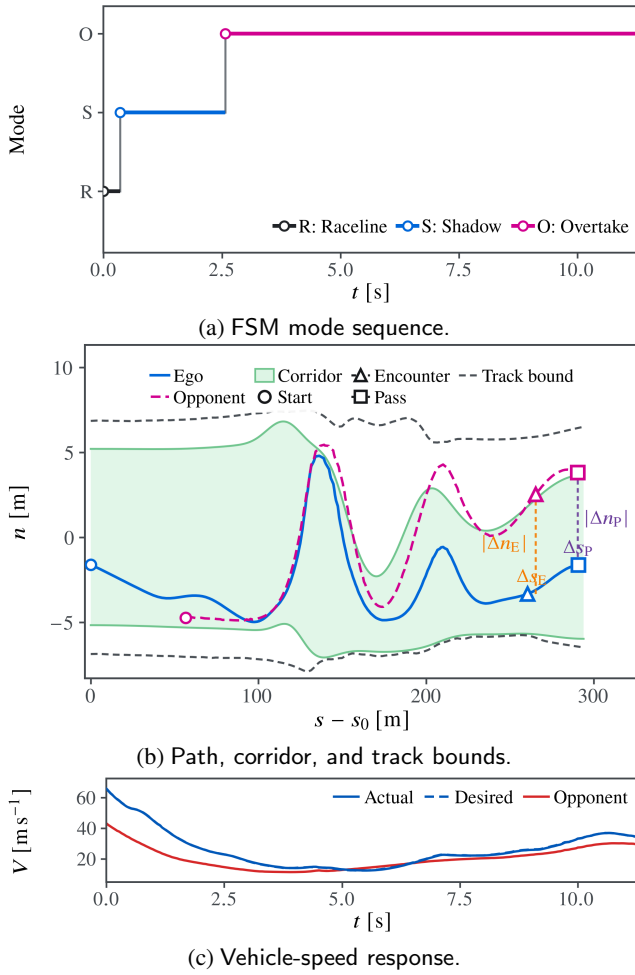


Figure 8: Tactical and speed evolution in Case II. (a) The FSM retains the selected passing side through the curved engagement. (b) The planned path and published corridor remain inside the varying physical track boundaries. (c) The actual speed tracks the desired braking and re-acceleration profile.

Case II: persistent overtake through a sharp-corner sequence

Case II combines an entry speed above 60 m s^{-1} with a longer interaction through successive curvature changes. Figure 8 shows that the FSM establishes SHADOW and then retains OVERTAKE through the curved engagement. As the physical track width changes, the carver continuously reshapes the locked-side corridor and the planned path remains within the available channel. The speed response follows the desired braking profile into the corner and the subsequent acceleration above 35 m s^{-1} , maintaining the passing opportunity throughout the sequence.

The simultaneous braking, turning, and relative-position change produces larger transient errors than in Case I (Fig. 9). Lateral and heading errors return toward zero after each curvature transition, speed tracking remains bounded during braking and re-acceleration, and the acceleration samples remain inside the tire envelope over the recorded speed range. Together, the two cases show that the learned

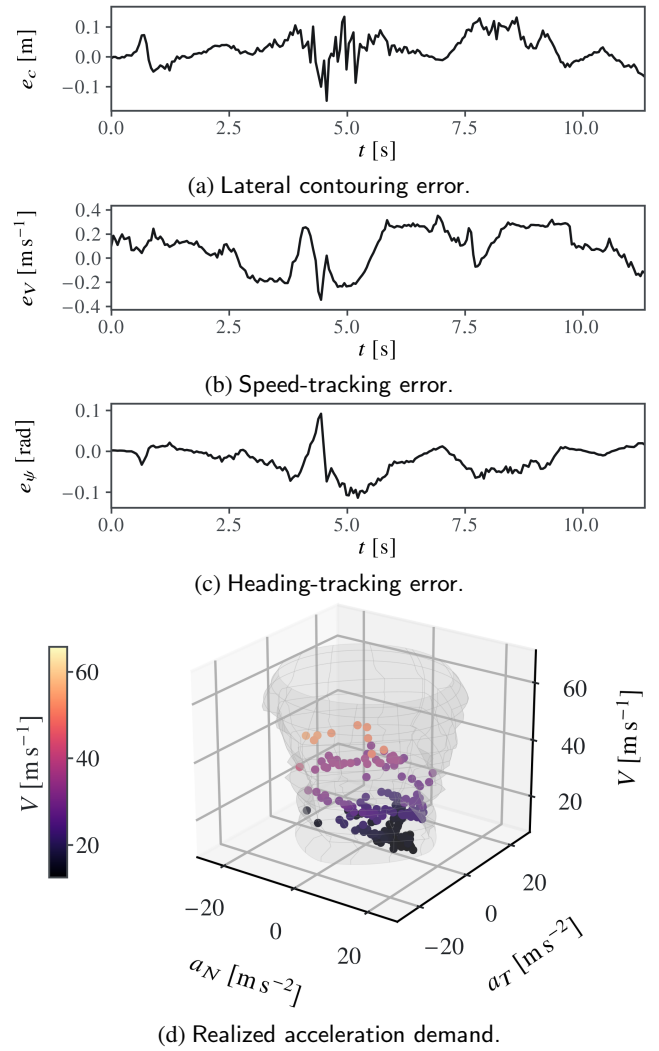


Figure 9: Closed-loop tracking and tire-demand response in Case II. (a–c) Tracking remains bounded through the prolonged curved interaction, including the braking, rotation, and exit phases. (d) All recorded acceleration samples remain inside the declared speed-dependent tire-demand envelope.

corridor remains persistent and executable during both a pass above 70 m s^{-1} and a constrained corner sequence.

Across 1000 Monte Carlo runs, the full method achieved an OSR of 94.7%, with collision and track-violation rates of 0.0%. The matched comparisons below quantify the contribution of the interactive tactical representation and its constituent modules.

5.3. Comparison with tactical baselines

All methods use the same plant, race-control interface, corridor projection, path and speed planners, and control module, with the tactical guidance module defining each comparison. *Pure Raceline* follows the nominal racing line. *Non-Interactive MPC* predicts the opponent independently of the ego plan and imposes obstacle constraints on the result [1, 3]. *Rule-based GT* uses a fixed-threshold state machine ($g_{\text{chase}} = 20 \text{ m}$, $g_{\text{ot}} = 8 \text{ m}$, and $g_{\text{abort}} = 35 \text{ m}$). *iLQGames* computes coupled local responses with iterative linear-quadratic game approximations [13]. Outputs from all

Table 1

Matched comparison over 1000 Monte Carlo closed-loop HIL runs per method. Arrows indicate the preferred direction.

Metric	Prop.	PR	NI-MPC	RB-GT	iLQG
OSR [%] $\uparrow$	94.7	38.2	28.5	51.6	63.5
OSR def. [pp] $\downarrow$	0.0	56.5	66.2	43.1	31.2
TTO [s] $\downarrow$	4.82	7.25	11.23	9.37	7.83
Min. gap [m] $\uparrow$	5.83	1.42	3.85	2.14	2.31
Contact [%] $\downarrow$	0.0	2.1	0.7	4.8	1.4
Mean TTC [s] $\uparrow$	8.52	3.21	6.92	5.64	5.73
Track viol. [%] $\downarrow$	0.0	0.0	0.0	0.0	0.0
PFR $\uparrow$	1.000	0.988	0.998	0.987	0.995

Notes: Prop., proposed method; PR, Pure Raceline; NI-MPC, Non-Interactive MPC; RB-GT, Rule-based GT; iLQG, iLQGames. OSR def. is the percentage-point deficit from Prop. Min. gap is the mean per-lap minimum inter-vehicle distance. PFR is the fraction of planning cycles with a feasible SOCP solution. Bold denotes the best observed value; equal track violation values are jointly bold.

baselines are projected onto the same side, speed-cap, and corridor interface, yielding a matched comparison of tactical reasoning.

The proposed tactical-corridor policy completed 94.7% of the laps, exceeding the strongest baseline, iLQGames, by 31.2 percentage points while increasing the mean minimum gap from 2.31 to 5.83 m and reducing the contact rate from 1.4% to 0.0% (Table 1). iLQGames achieved the strongest baseline result, showing the value of coupled interaction reasoning, whereas the additional conversion of that reasoning into a persistent feasible region markedly increased successful closed-loop realization. The PFR of 1.000 further shows that the published corridors consistently supplied usable domains to the common planner across the evaluated laps.

5.4. Ablation study

Four ablations isolate the proposed tactical components while retaining the same execution layer. *Without game guidance* removes both the Stackelberg response-set action prior and the action-conditioned robust-value supervision. *Without FSM* bypasses hysteretic mode and side locking, allowing the policy to reshape the corridor on every update. *Without learned corridor* replaces the continuous width, clearance, lateral-bias, and speed-cap outputs by fixed nominal values. The final variant replaces TQC with Soft Actor-Critic (SAC) while preserving the network, replay, reward, and other training settings.

The ablations trace the performance gain to the three stages of the proposed interface (Table 2). Fixing the corridor produced the largest effect, reducing OSR by 18.6 percentage points, increasing the contact rate from 0.0% to 3.1%, and yielding the lowest PFR; adaptive feasible-set shaping therefore determines whether a usable passing channel is formed. Removing the FSM reduced OSR by 12.4 points and raised the contact rate to 1.2%, showing the importance of preserving that channel across successive updates. Removing game guidance reduced OSR by 9.3 points and increased TTO by 1.56 s, linking response-set supervision to opportunity timing and clearance. Replacing TQC with SAC produced a 6.2-point OSR reduction and

Table 2

Component ablations over 1000 Monte Carlo closed-loop HIL runs per variant. Each variant removes or replaces one tactical component. Arrows indicate the preferred direction.

Metric	Full	No-GG	No-FSM	Fixed-C	SAC
OSR [%] $\uparrow$	94.7	85.4	82.3	76.1	88.5
OSR loss [pp] $\downarrow$	0.0	9.3	12.4	18.6	6.2
TTO [s] $\downarrow$	4.82	6.38	6.71	8.94	5.53
Contact [%] $\downarrow$	0.0	0.9	1.2	3.1	0.5
Min. gap [m] $\uparrow$	5.83	2.86	2.17	1.05	3.94
Mean TTC [s] $\uparrow$	8.52	4.93	4.31	2.86	6.17
PFR $\uparrow$	1.000	0.983	0.974	0.961	0.992

Notes: Full, full method; No-GG, without game guidance; No-FSM, without the finite-state machine; Fixed-C, fixed corridor. OSR loss is the percentage-point difference from Full. PFR is the fraction of planning cycles with a feasible SOCP solution. Bold denotes the best observed value.

Table 3

Sensitivity results over 1000 Monte Carlo runs per setting.

Setting	OSR [%] $\uparrow$	TTO [s] $\downarrow$	$g_{\min}$ [m] $\uparrow$	Contact [%] $\downarrow$	PFR $\uparrow$
Nominal	94.3	4.82	5.83	0.0	0.9990
$\epsilon_{BR} = 0.025$	92.4	4.71	4.55	0.6	0.9926
$\epsilon_{BR} = 0.10$	93.0	5.70	6.22	0.1	0.9980
$\lambda_p = 0$	90.8	5.70	4.28	0.7	0.9916
$\lambda_p = 2$	93.1	5.29	5.51	0.3	0.9968
$\lambda_g = 0$	92.3	5.31	4.85	0.5	0.9938
$\lambda_g = 2$	93.4	5.21	5.78	0.1	0.9980
$\lambda_p = \lambda_g = 0$	85.4	6.38	2.86	0.9	0.9830

a 0.5% contact rate, indicating an additional benefit from truncating optimistic return quantiles.

5.5. Sensitivity analysis of hyperparameters

Each setting was retrained and evaluated over 1000 Monte Carlo closed-loop runs; ϵ_{BR} changes also regenerated the teacher labels. Table 3 shows that halving ϵ_{BR} shortened TTO from 4.82 to 4.71 s, but reduced the minimum gap from 5.83 to 4.55 m and raised the contact rate to 0.6%. Doubling it instead increased the gap to 6.22 m with a 0.1% contact rate, but lengthened TTO to 5.70 s, revealing the intended efficiency–robustness trade-off. Removing the action prior or robust-value supervision reduced OSR to 90.8% and 92.3%, respectively. Joint removal produced the largest OSR loss of 8.9 percentage points, whereas doubling either loss weight retained OSR above 93% and PFR above 0.996.

5.6. Algorithm evaluation on unseen circuits

The Yas Marina-trained tactical–execution stack was evaluated on the unseen Imola and Suzuka circuits using the same policy and execution parameters to examine whether the track-relative feasible-set representation transfers to new geometries. Figures 10 and 11 each localize four overtaking events and pair the corresponding SHADOW and OVERTAKE simulator views. Imola introduces separated braking zones and curved approaches, whereas Suzuka combines high-speed direction changes, tight turns, and long acceleration sections. These geometries exercise the track-relative observation, persistent side commitment, and corridor carver beyond the circuit used for training.

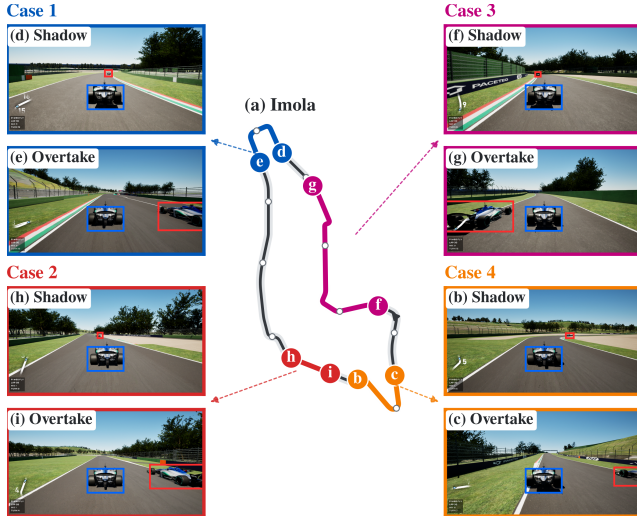


Figure 10: Representative overtaking events on the unseen Imola circuit. The track map localizes four events, with paired SHADOW and OVERTAKE simulator views at each location.

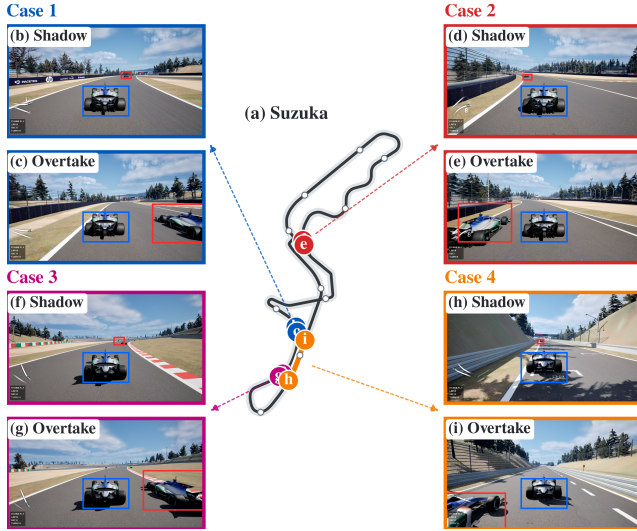


Figure 11: Representative overtaking events on the unseen Suzuka circuit. Four track locations connect the approach and committed passing phases across distinct local geometries.

Table 4

Aggregate performance over 1000 Monte Carlo runs on each unseen circuit.

Metric	Imola	Suzuka
OSR [%]	91.8	88.9
TTO [s]	5.19	4.68
Minimum gap [m]	5.46	5.05
Mean TTC [s]	8.07	7.45
Contact rate [%]	0.2	0.5
Track-violation rate [%]	0.0	0.1
PFR	0.9968	0.9934

Notes: Continuous entries are Monte Carlo means; TTO is averaged over successful overtakes. Binary outcomes are reported as percentages.

Across the eight illustrated events, the tactical state progresses from approach to committed passing on both straight and curved sections, while the corridor remains aligned with

Table 1

Principal algorithm and closed-loop evaluation settings.

Component	Parameter	Value
Vehicle geometry	$m; l_f, l_r$	742 kg; 1.75, 1.35 m
Tire curves	$F_{z,0}; C_x, C_y; E_x, E_y$	2079.72 N; 1.6144, 1.5998; 0, 0
Tire curves	$\mu_x, \mu_y; s_x, s_y; q_{x,pk}, q_{y,pk}$	2.2345, 1.9745; -0.2555, -0.4787; 0.1296, 0.140
Tire envelope	construction; online evaluation	quasi-steady-state map (2); speed interpolation
Time scales	$T_{tac}; T_c$; station spacing	0.02 s; 0.01 s; 2 m
Response-set teacher	Sobol candidates per side; ϵ_{BR}	64; 0.05; 0.99
Action projection	$\gamma_g; q_{neu}$	0.5 m; 0.25
FSM	N_{post} ; stale-reference limit	3 updates; 1 revaluated cycle
TQC	critics; N_g ; retained K ; γ	2; 25; 46; 0.99
TQC guidance	$\lambda_p; \lambda_g$; prior metric	1; 1; identity after action normalization
Corridor	preview; stations; L_{min}^{corr} ; β	300 m; 150; 3 m; 0.2
Frenet health	$\epsilon_s; \epsilon_{max}; v_s$	0.05; 1.30 rad; 1 m s ⁻¹
Path health	iterations; $(\epsilon_{dyn}, \epsilon_{cor})$	5; $(10^{-3}, 10^{-3})$ m
Speed profile	$H_s; V_{min}, v_{x,max}; \epsilon_\rho$	150; 15, 90 m s ⁻¹ ; 10 ⁻³
RC-NMPC	steps; $T_c; \delta_{max}; \delta_{max}^*$; ΔV_{max}	15; 0.01 s; 0.35 rad; 1.0 rad s ⁻¹ ; 0.5 m s ⁻¹
RC-NMPC weights	$q_{\Delta\delta}, q_c, q_\psi; R_v$	2, 20, 10; diag(1, 0.5)

the local track geometry. The aggregate outcomes in Table 4 quantify this cross-circuit transfer.

6. Conclusion

This paper addressed the persistence and executability of opponent-aware overtaking decisions under repeated replanning. The proposed method represents the decision as a side-committed, time-varying tactical feasible region. A robust Stackelberg response-set teacher supervises where and when to form this region, TQC distills the supervision, and the hysteretic state machine with deterministic carving maintains it as the encounter evolves. A common sequential-SOCP and RC-NMPC executor then converts the published corridor into closed-loop path-speed motion. On the target RGS-8805GC computer in a dual-CAN EAV24 HIL loop, the method achieved 94.7% overtaking success over 1000 Monte Carlo closed-loop simulations and exceeded four matched baselines by 31.2–66.2 percentage points. The fixed-corridor and no-FSM ablations produced the two largest performance losses, directly linking adaptive feasible-set generation and temporal persistence to successful execution. These results establish tactical-corridor learning as an effective interface between interaction reasoning and constrained vehicle planning within state-informed, two-vehicle autonomous racing.

A. Algorithm and Evaluation Parameters

This appendix reports the numerical settings and normalized reward features needed to reproduce the algorithmic logic.

Table 2

Tactical-policy bounds, reward weights, and training settings. Symbols follow Eqs. (6) and (12).

Group	Setting
Corridor and speed	$L^{\text{cap}}, R^{\text{cap}} \in [1.5, 6.0] \text{ m}; d_{\text{safe}} \in [0.2, 0.8] \text{ m}; V_{\text{cap}}^{\text{req}} \in [15, 90] \text{ m s}^{-1}$
Gap thresholds	$g_{\text{chase}} \in [18, 24] \text{ m}; g_{\text{ot}} \in [16, 24] \text{ m}; g_{\text{abort}} \in [28, 45] \text{ m}$
Bias, recovery, side	$b_{\text{lat}} \in [-1.5, 1.5] \text{ m}; w_{\text{safe}} \in [1, 5] \text{ m}; q_{\text{side}} \in [-1, 1]$
Reward weights	$\mathbf{w} = (30, -20, -100, -2, -5, 3, -0.1, -0.5, 0.5, 0.25)$, ordered as in Eq. (12)
Optimization	learning rate 3×10^{-4} ; replay capacity 10^5 ; batch size 256; target update 5×10^{-3}
Networks	independent actor encoder 51 $\rightarrow$ 256 $\rightarrow$ 256; shared critic state-action encoder; two quantile heads; training-only robust-value head
Projection gradients	piecewise Π_A ; exact Π_{tire} forward pass with STE for actor/prior updates; detached Bellman target
Software	Python 3.10.0; SB3 2.8.0; PyTorch 2.4.0+cu124; Gymnasium 1.2.3

$$\begin{aligned}
\phi_{\text{ot}} &= \mathbb{1}[m_t = \text{OVERTAKE}, m_{t+1} = \text{RACELINE}, P_t], \\
\phi_{\text{ovb}} &= \phi_{\text{ot}} \text{ with roles exchanged,} \\
\phi_{\text{col}} &= \mathbb{1}[\text{contact} \vee \text{offtrack} \vee \text{comm_loss}], \\
\phi_{\text{lat}} &= \mathbb{1}[m_t = \text{RACELINE}] \text{clip}\left(\frac{n_e - n_{t1}}{W_{\text{trk}}/2}, -1, 1\right)^2, \\
\phi_{\text{crush}} &= \mathbb{1}[|g_t| < g_{\text{ot}}] \left(\frac{[\rho_{\text{exc}} - |\Delta n|]_+}{\rho_{\text{exc}}}\right)^2, \\
\phi_{\text{app}} &= \mathbb{1}[0 < g_t < g_{\text{chase}}] \text{clip}\left(\frac{g_{t-1} - g_t}{g_{\text{chase}}}, 0, 1\right), \\
\phi_{\text{sm}} &= \min\left\{1, \frac{\|N_a(\mathbf{a}_t^F) - N_a(\mathbf{a}_{t-1}^F)\|_2^2}{10}\right\}, \\
\phi_{\text{gs}} &= \min\left\{1, \frac{1}{2N_{\text{sta}}} \sum_k \sum_{\pm} \left(\frac{n_{k,t}^{\pm} - n_{k,t-1}^{\pm}}{W_{\text{trk},k}}\right)^2\right\}, \\
\phi_{\text{shs}} &= \mathbb{1}[|g_t| < g_{\text{ot}}, |\Delta n| \geq \rho_{\text{exc}}], \\
\phi_{\text{wide}} &= \frac{1}{N_{\text{sta}}} \sum_k \text{clip}\left(\frac{n_{k,t}^+ - n_{k,t}^- - L_{\text{min}}^{\text{corr}}}{W_{\text{trk},k} - L_{\text{min}}^{\text{corr}}}, 0, 1\right).
\end{aligned} \tag{20}$$

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